

Expense Tracker

AI-Built Personal Finance Tracker | AI Champion Showcase | May 2026

Why I Built This

I wanted to visualize where my money goes each month. A Google Sheet works for data entry but can't spot trends, compare months, or track budgets. The bigger insight: if I struggle with this as a data person, how do small business owners (restaurants, retail shops) manage? They don't have data teams.

The Stack (Built Entirely with Claude AI)

[DB]

Supabase

PostgreSQL Cloud DB

[API]

FastAPI

Python REST API

[T]

dbt

Data Transforms

Where AI Fits In

AI in HOW it was built

- Schema + 4 tables designed by AI
- 14 API endpoints auto-generated
- 5 dbt models with AEST timezone
- 6+ errors debugged live by AI
- Zero manual coding required

AI in the PRODUCT (Vision)

Phase 2: AI CSV cleaner

- Auto-map columns to schema
- Detect categories from desc

Phase 3: AI dashboard builder

- Customer uploads Excel -> BI
- No data engineer needed

My Data Engineering & AI Journey

[MS] Microsoft Fabric / Synapse

- TEEG CID project: production data
- 60+ Synapse notebooks
- Bronze -> Silver Lakehouses
- CI/CD pipelines (Azure DevOps)
- Power BI semantic models
- AEST timezone conversions

[AI] Databricks AI Agents

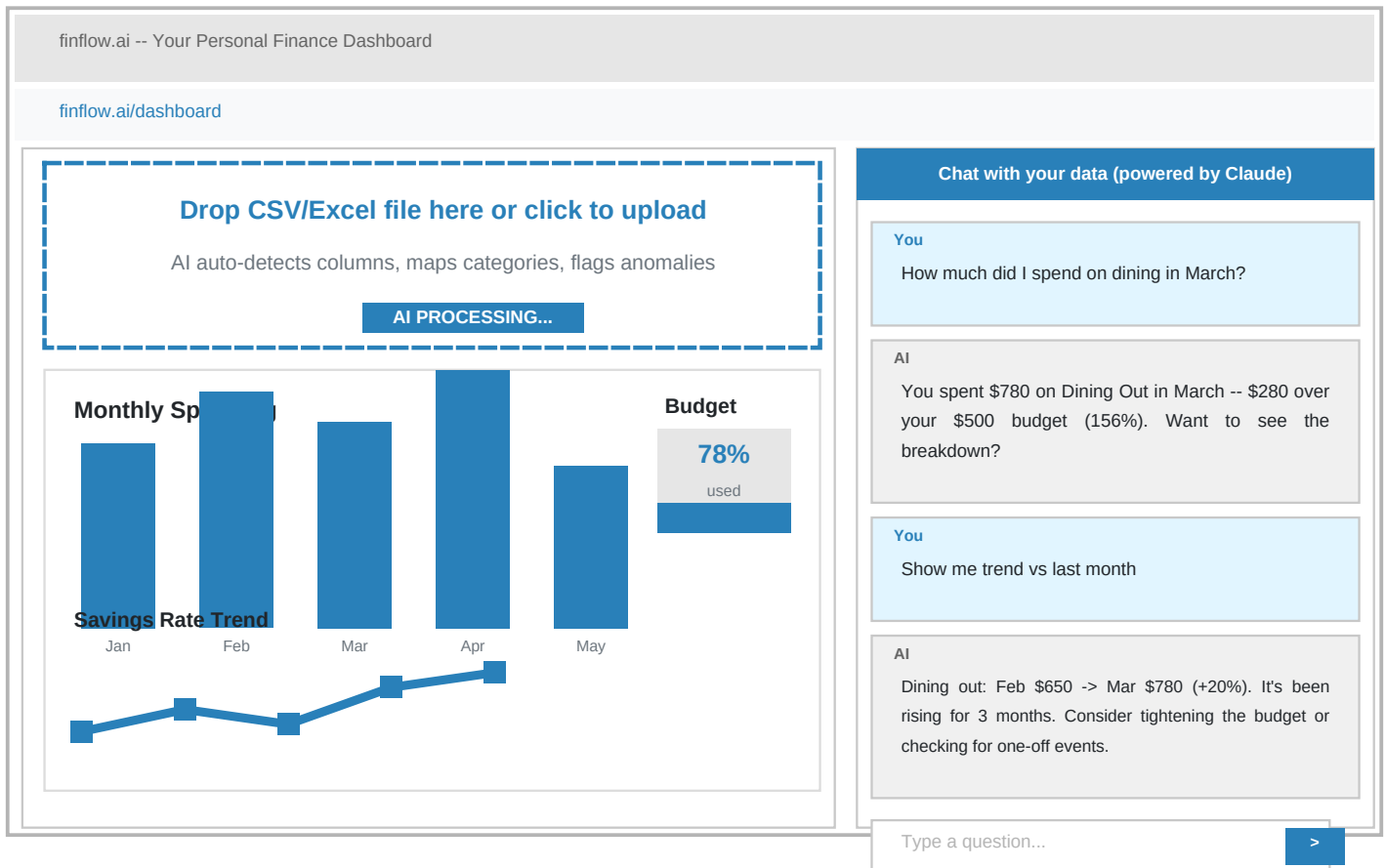
- Databricks AI Days conference
- Built RAG agents (tool-calling)
- MLflow ResponsesAgent + UC funcs
- Vector search for documents
- Agent evaluation + deployment
- Agent Bricks (no-code UI)

Key Takeaways

1. AI democratizes data engineering -- non-developers can build production-grade data stacks.
2. Real tools, real stack -- Supabase, dbt, FastAPI, Power BI. Same tools enterprises use.
3. Start with personal pain, scale to product -- best SaaS ideas come from solving your own problem.
4. AI is the co-builder -- Claude designed, coded, debugged, and explained every layer.

The Product Vision

What the customer sees -- and what powers it behind the scenes



Three Modes of AI Interaction

[1] BUILD (now)

- Claude Code in VS Code
- Designs schema + API
- Writes dbt models
- Debugs errors live
- You: 'Add a table...'

[2] OPERATE (future)

- Web chat (customer)
- Claude API + tool calls
- Queries Supabase
- Answers in natural lang
- 'How's my budget?'

[3] VISUALIZE (future)

- Power BI embedded
- Real-time dashboards
- Auto-refresh on upload
- Customer sees insights
- No data skills needed

Architecture: What Powers the Chat



agent_tools.py -- The Bridge Between Chat and Data

```
def get_monthly_summary(year, month):  
    """Return spending by category for a given month."""  
    # Queries Supabase v_monthly_summary  
  
def compare_months(month1, month2):  
    """Compare spending between two months side by side."  
    Built with Claude AI | From personal tracker -> AI SaaS
```